Mechanisms of Social Status Regulation as Free–Energy Minimising Loops

Compiled for AI Safety Collaborators

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Abstract

Human status dynamics appear to be governed by a portfolio of partially independent optimisation loops, each minimising a distinct component of variational free energy. We propose a compact taxonomy—from reptilian dominance reflexes to language–mediated reputation markets—and specify, for each loop, (i) the free–energy term measured, (ii) the neuroanatomical registers and actuators, and (iii) a plausible micro–mechanism that yields a prediction–error (PE) signal with neuromodulatory precision–gating. We articulate an internal/external symmetry (interoceptive threat vs. exteroceptive perimeter defence; somatic repair vs. niche construction; etc.), derive perturbation predictions, and outline how these principles transfer to opaque–agent forensics and brain–like AI benchmarks.

1 Background

Under the free–energy principle (FEP) [1], adaptive systems minimise expected surprise by building predictive models and selecting actions that make sensory data conform to those models. Status regulation can be cast as a set of social free–energy ledgers: uncertainty about threat, coalition membership, norm adherence, prestige sources, and reputational standing. Evolutionary continuity suggests layered solutions: brainstem circuits for dyadic dominance, limbic coalition bookkeeping, and neocortical narrative broadcast [2, 3, 4].

2 Loops, ledgers, and evolutionary stratification

Table 1 orders status loops from older to newer and states, for each, the free–energy target F, registers, actuators, and neuromodulators that set PE precision.

3 How circuits *measure* the ledgers

Across loops, a common micro–architecture appears: deep pyramidal cells carry priors; superficial cells compute residuals; neuromodulators scale precision (inverse variance). Concretely:

- **Dominance threat:** BLA sends expected threat to VMHvl; MeA relays opponent cues. Subtractive interneurons in VMHvl produce a PE that recruits PAG; raphe 5–HT sets risk tolerance, testosterone gates plasticity [5, 2].
- **Coalition entropy:** TPJ/mPFC encode a sparse triadic graph prior. pSTS supplies live affiliative cues. Error neurons in TPJ fire when link likelihood deviates, with OXT increasing gain (precision) on in–group edges [8, 9].
- **Prestige:** Mirror system (IFG/pSTS) hosts a predictive motor program of the model; cerebellum/PPC compute kinematic mismatch; VTA DA encodes epistemic PE that reweights imitation [10, 7].
- **Norms:** vmPFC stores rule templates; ACC compares allowed action vs. observed; an aversive PE drives dlPFC→BG sanction. Relief upon resolution corresponds to a drop in F_{norm} and engages μ–opioid [13, 14].

Age	Loop	Free–energy term F	Registers (priors / evidence)	Actuators	Precision gate
1	Dominance–submission	$F_{\text{threat}} = \Pr(\text{injury} \mid \pi)$	BLA prior $\leftrightarrow$ VMHvl, MeA evidence	PAG columns (fight/flight), spinal patterns	5–HT, T [5, 6]
2	Display–assessment	$F_{\text{signal}} = \lvert \hat{s}_{\text{self}} - s_{\text{peer}} \rvert$	Ventral striatum template $\leftrightarrow$ auditory/visual cortex	Pattern generators (laryngeal, facial)	DA [7]
3	Coalition entropy	$H(C_{ij})$	TPJ/mPFC graph prior $\leftrightarrow$ STS cues	Grooming/approach, alliance choice	OXT [8, 9]
4	Prestige–biased learning	$F_{\text{ep}}^{\text{social}} = \mathbb{E}\,\lvert o - \hat{o}_{\text{model}} \rvert$	IFG+pSTS mirror prior $\leftrightarrow$ PPC/cerebellar mismatch	Imitation weighting, teaching	DA [10, 11]
5	Norm enforcement	$F_{\text{norm}} = \Pr(\text{rule breach})$	vmPFC/hippocampal rule prior $\leftrightarrow$ ACC/insula error	dlPFC$\rightarrow$BG punishment policy	μ–opioid, NA [12, 13, 14]
6	Reputation broadcast	$F_{\text{rep}} = \lvert \hat{r}_i - r_i \rvert$	DMN (PCC, temp. pole) prior $\leftrightarrow$ STG language input	Narrative update, signalling	DA/ OXT [4, 3]

Table 1: Status loops from evolutionarily older (1) to newer (6). Abbreviations: BLA basolateral amygdala; VMHvl ventromedial hypothalamus (ventrolateral); MeA medial amygdala; TPJ temporoparietal junction; STS superior temporal sulcus; IFG inferior frontal gyrus; PPC posterior parietal cortex; dlPFC dorsolateral prefrontal cortex; BG basal ganglia; DMN default mode network; NA noradrenaline; OXT oxytocin; DA dopamine; 5–HT serotonin; T testosterone.

- **Reputation:** DMN maintains a semantic centrality prior; language cortex updates evidence; ventral striatum emits a value PE proportional to change in expected future cooperation [4].

4 Internal $\leftrightarrow$ external symmetry

Each interoceptive ledger has an exteroceptive twin: internal threat (amygdala/PAG) vs. external perimeter defence (SC–guided vigilance); internal somatic repair vs. external niche construction; internal norm adherence (guilt) vs. external sanction (punishment). This symmetry follows from FEP at two scales: body maintenance and niche shaping.

5 Perturbation predictions

TMS to dlPFC (norm loop)

Continuous theta–burst TMS over right dlPFC should attenuate third–party punishment without reducing ACC mismatch (N200/P300). Behaviourally, acceptance of unfair splits rises; sanction spending falls [13]. Relief signals (opioid) diminish until excitability recovers.

Serotonergic shift (dominance loop)

An acute SSRI dose (increased 5–HT) raises the submit threshold and reduces aggressive approach under status threat; the PE amplitude in VMHvl drops for identical opponent cues [6].

6 Consequences for agency and consciousness

Status loops feed into valuation–efference circuits: changes in F_{threat} and $H(C_{ij})$ directly alter discounting, exploration, and social policy selection [15]. Global broadcast and metacognition render these computations available to report, shaping conscious feelings of pride, shame, outrage. Miscalibrated precision (e.g., high OXT precision on in–group edges) can lower social free energy locally while increasing global risk (out–group hostility), a classical alignment pitfall for artificial agents.

7 Opaque–agent forensics: low–sample discovery

Borrowing from lesion logic, we recommend four probes per discovered module: (i) topology screen (hub vs. chain), (ii) one–bit poke (rebound gain $g > 1$ indicates homeostasis), (iii) 30 s behavioural battery (rank cue, norm breach, prestige honeypot), and (iv) power snapshot. A triple–hit (structure+PE+behaviour) labels a status loop without long recordings.

8 Outlook

The status stack recapitulates vertebrate evolution while exploiting neocortical broadcast. Incorporating these ledgers into AI benchmarks should surface prestige–seeking, coalition maintenance, and norm enforcement as distinct optimisation targets, enabling targeted "cooling" or constraint of each loop.

Limitations. Citations are indicative; multiple parallel routes can implement the same PE computation. Quantitative priors (precision) are context–dependent and ethically sensitive to manipulate in humans.

References

[1] K. Friston (2010). The free-energy principle: a unified brain theory? *Nat. Rev. Neurosci.* 11:127–138.

[2] R. Sapolsky (2004). Social status and health in humans and other animals. *Annu. Rev. Anthropol.* 33:393–418.

[3] J. Henrich & F. Gil-White (2001). The evolution of prestige. *Evol. Hum. Behav.* 22:165–196.

[4] C. F. Zink et al. (2008). Know your place: neural processing of social hierarchy. *Neuron* 58:273–283.

[5] D. Lin et al. (2011). Functional identification of an aggression locus in the mouse hypothalamus. *Nature* 470:221–226.

[6] M. J. Raleigh et al. (1991). Serotonergic mechanisms in vervet dominance. *Brain Res.* 559:181–190.

[7] W. Schultz et al. (1997). A neural substrate of prediction and reward. *Science* 275:1593–1599.

[8] J. Scholz et al. (2016). rTPJ in attention and social cognition: combined fMRI–TMS. *Cereb. Cortex* 26:4282–4291.

[9] M. Kosfeld et al. (2005). Oxytocin increases trust. *Nature* 435:673–676.

[10] G. Rizzolatti & L. Craighero (2004). The mirror-neuron system. *Annu. Rev. Neurosci.* 27:169–192.

[11] M. Chudek & J. Henrich (2011). Culture–gene coevolution, prestige and conformity. *Phil. Trans. R. Soc. B* 366:1149–1158.

[12] E. Fehr & S. Gächter (2002). Altruistic punishment. *Nature* 415:137–140.

[13] D. Knoch et al. (2006). Disruption of right dlPFC decreases norm compliance. *Science* 314:829–832.

[14] N. I. Eisenberger et al. (2003). Does rejection hurt? *Science* 302:290–292.

[15] N. D. Daw et al. (2006). Cortical substrates for model-based vs. model-free learning. *Nat. Neurosci.* 8:1704–1711.